

Homework 6

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Contents

Problem 1	2
Summary	2
Explanation	2
Output	3

Problem 1

Summary

The following model is a SARIMA(1,1,0)x(2,1,2)[10] on X_t .

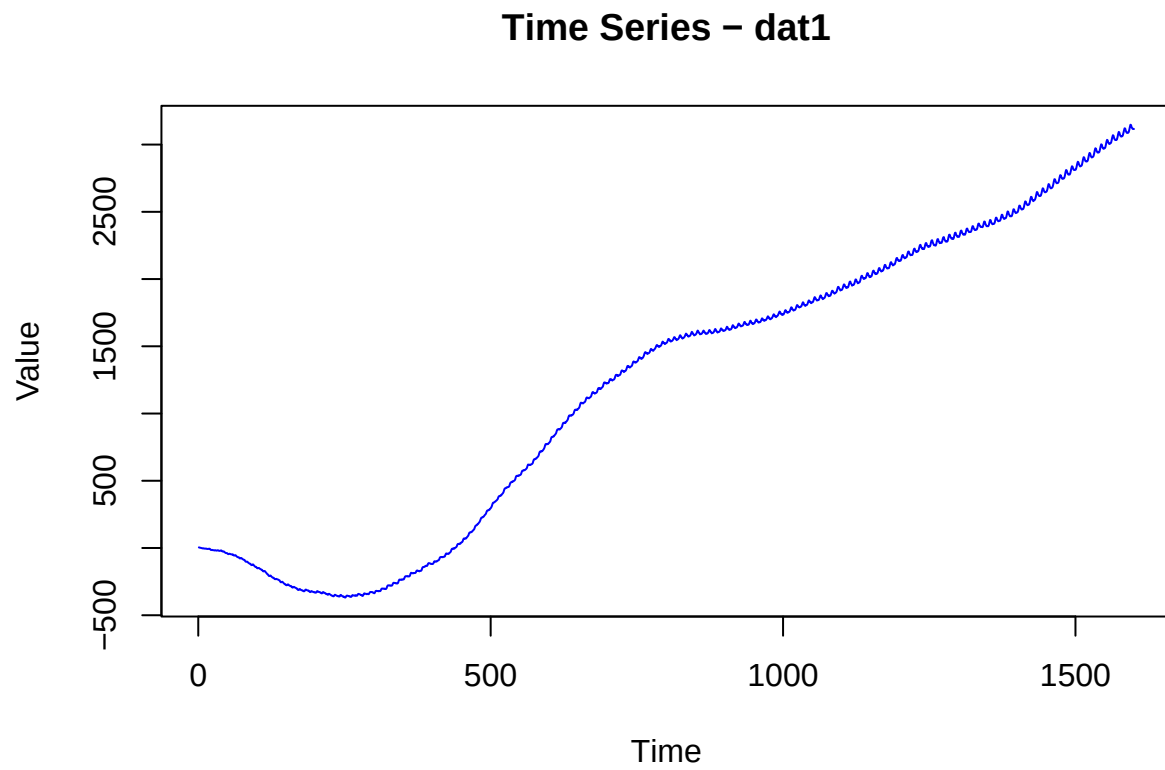
```
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.5104 0.0216  23.6057  0.0000
## sar1     -0.0161 0.0754  -0.2129  0.8314
## sar2      0.0087 0.0507   0.1722  0.8633
## sma1     -0.8075 0.0712 -11.3408  0.0000
## sma2      0.4312 0.0390  11.0604  0.0000
##
## sigma^2 estimated as 1.004128 on 1584 degrees of freedom
##
## AIC = 2.854927  AICc = 2.854951  BIC = 2.875207
##
##           Length Class  Mode
## coef           5  -none- numeric
## sigma2          1  -none- numeric
## var.coef       25  -none- numeric
## mask           5  -none- logical
## loglik          1  -none- numeric
## aic             1  -none- numeric
## arma           7  -none- numeric
## residuals 1600   ts      numeric
## call           8  -none- call
## series          1  -none- character
## code            1  -none- numeric
## n.cond          1  -none- numeric
## nobis           1  -none- numeric
## model          10  -none- list
```

Explanation

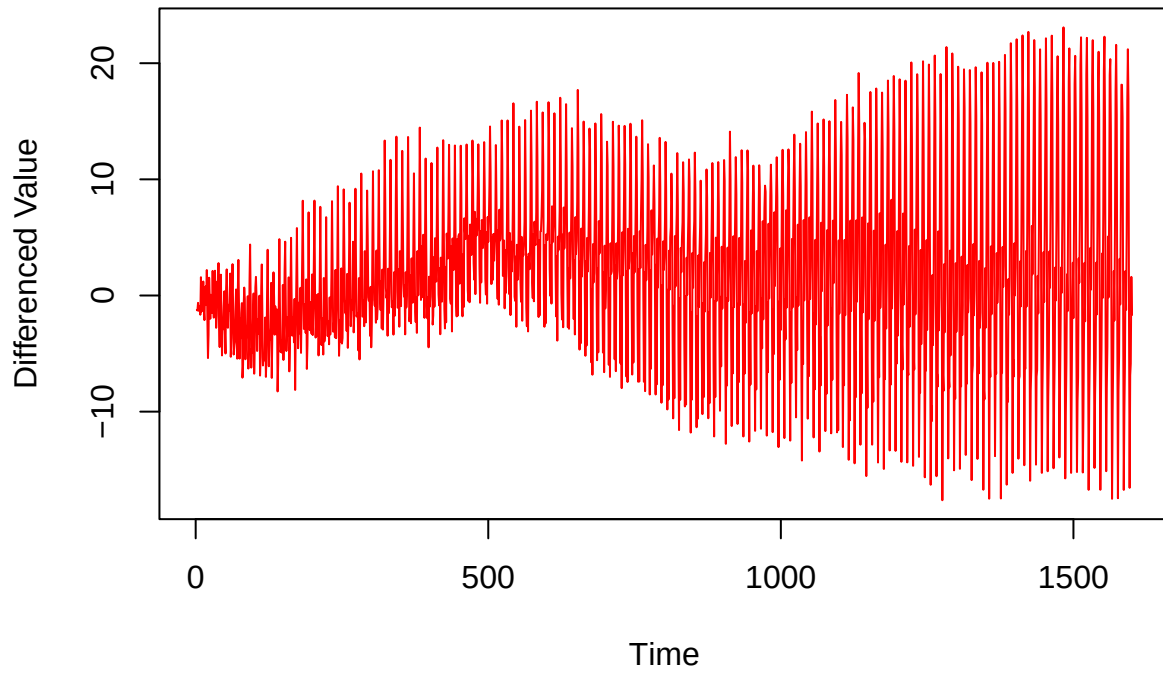
Looking at the initial plot of this series, X_t very clearly isn't stationary. The plot of the first difference of the series makes it clearer that there is also expanding variance as time increases, so a Box-Cox transformation was performed. Of the transformations plotted, the one in which $\lambda = 0.125$ looks the most stationary, so it was used for further analysis. However, when models were fit using this transformation, the models all showed extremely heavy tails in the normal QQ plot, so the original data without transformation was used. The ACF of X_t shows clear seasonality with period 10. After taking a seasonal difference with $s=10$, the ACF appears to tail off both seasonally and non seasonally, and the PACF appears to cut off abruptly after non seasonal lag 1 and seasonal lag 1, so the initial model fit was a SARIMA(1,1,0)x(1,1,0)[10]. This model showed three significant lags $h = 10, 20$, and 30 in the ACF of the residuals and the LBP showed cumulative autocorrelation that was unaccounted for at all lags, so a different model was fit with three additional terms. None of these four models had any concerningly significant lags in the ACFs of their residuals, and their LBPs had 2-3 lags below the threshold but were much better than that of the initial SARIMA(1,1,0)x(1,1,0)[10] model. Of these four models, the SARIMA(1,1,0)x(1,1,3)[10] and the SARIMA(1,1,0)x(2,1,2)[10] had the best AIC, and the SARIMA(1,1,0)x(2,1,2)[10] had fewer insignificant parameters at the $p < 0.01$ level so it was used as the final model.

Output

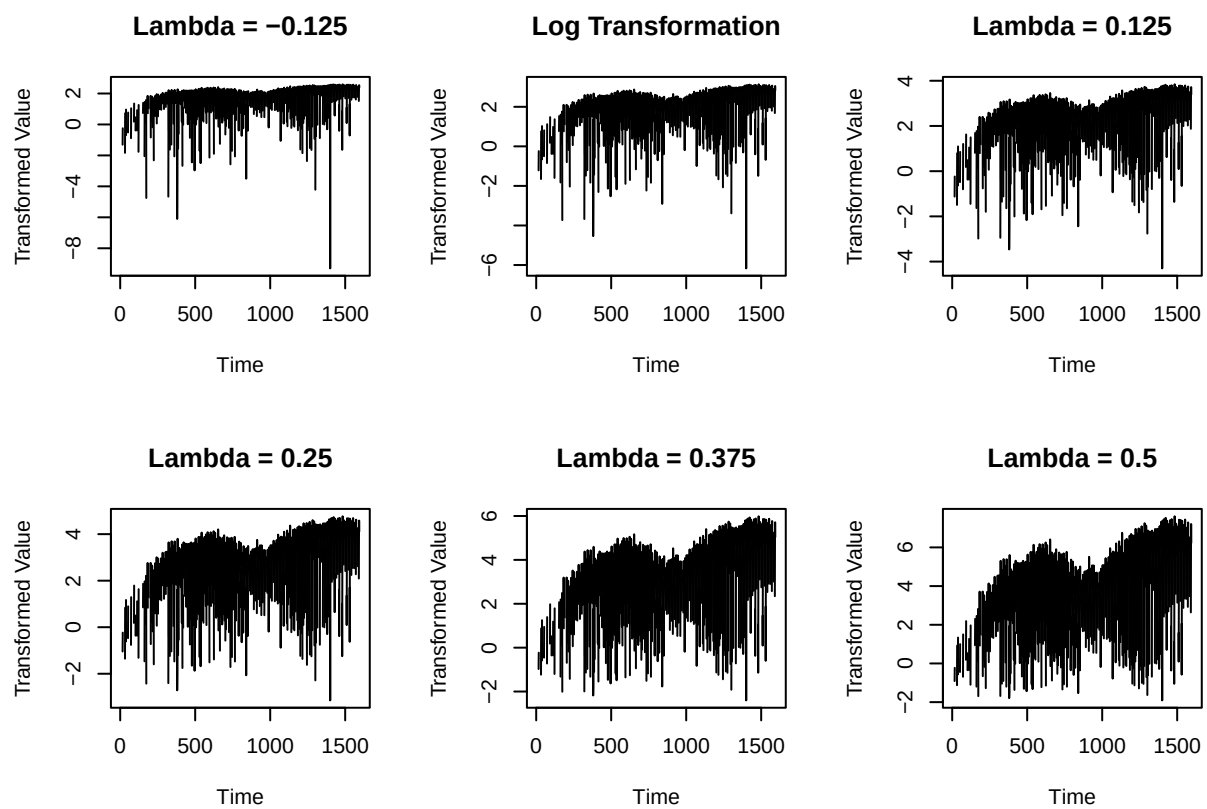
Plots of unadjusted and differenced series X_t .



Time Series – dat1 (Differenced)

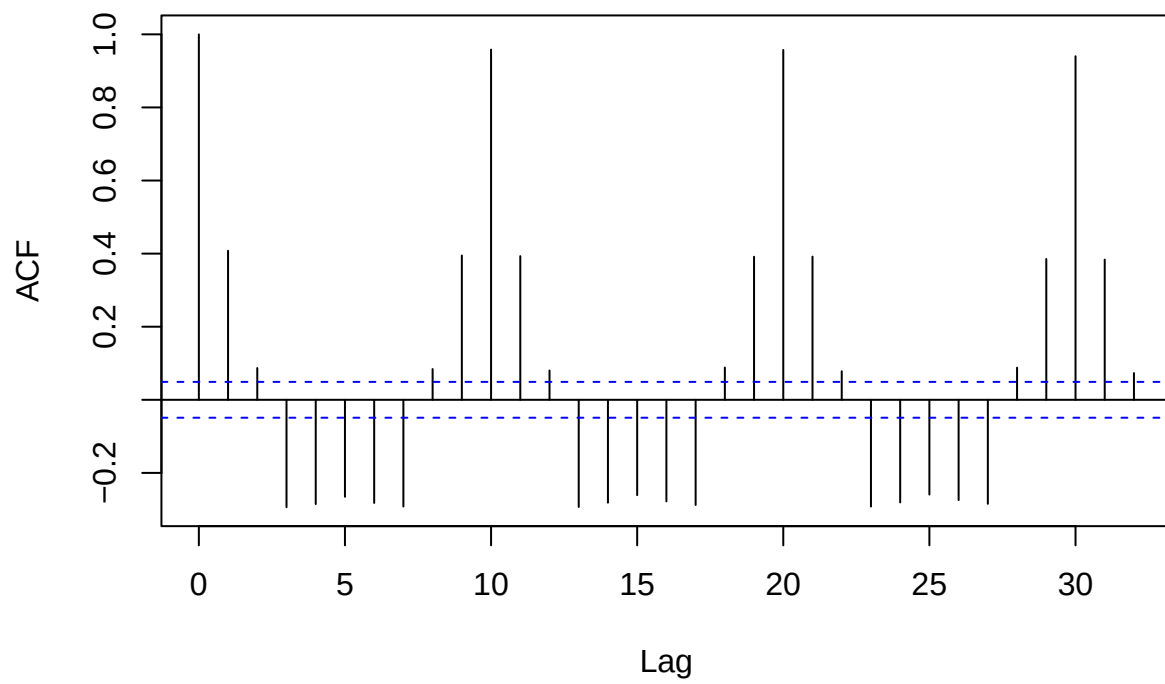


Plot of Box-Cox transformations of the differenced X_t with varying λ (no transformations ended up being used due to heavy tails in Normal QQ plot of models using transformed data):

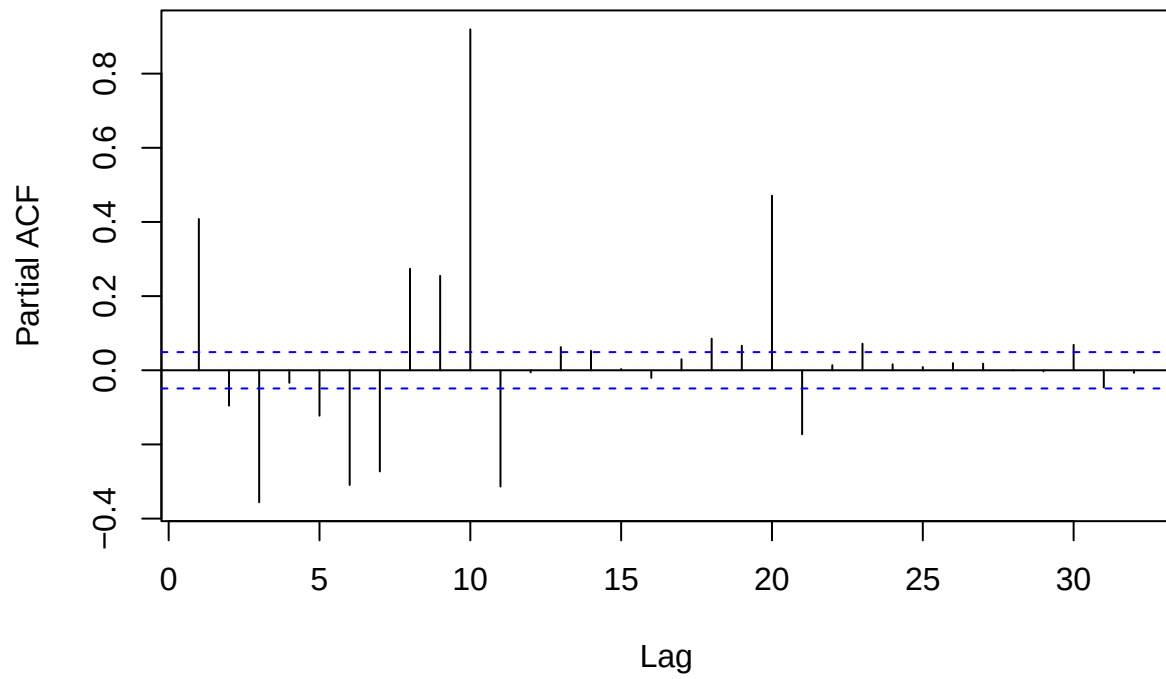


ACF and PACF of X_t

ACF of dat1_diff (Lambda = 0.125)

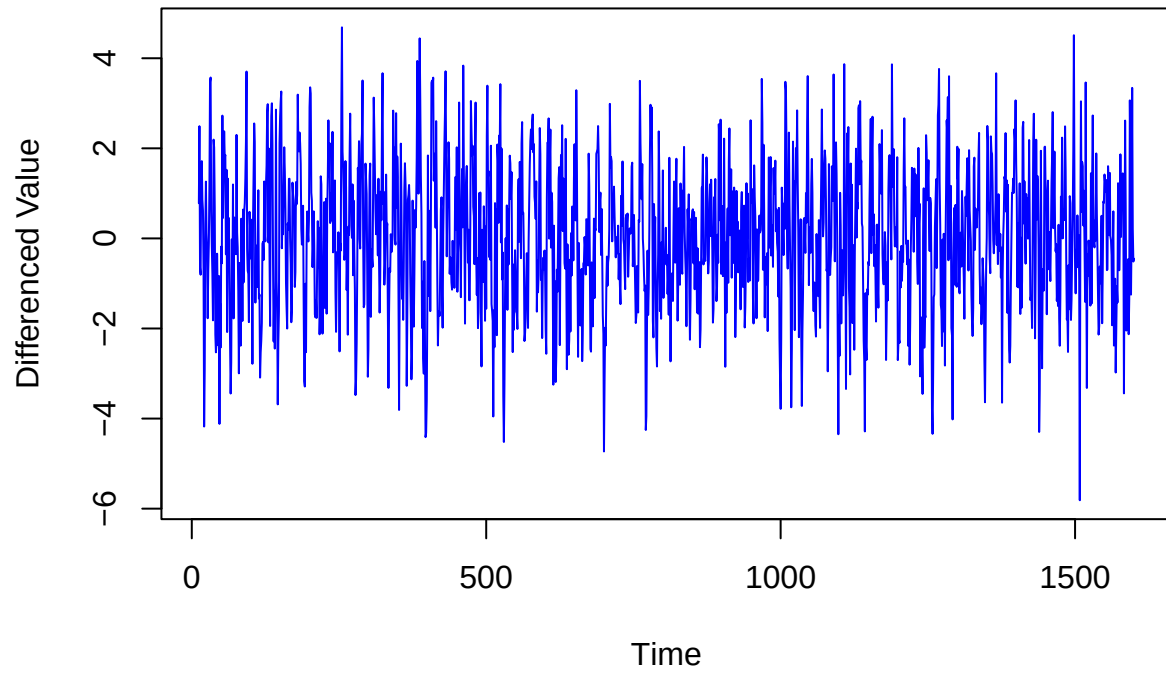


PACF of dat1_diff (Lambda = 0.125)

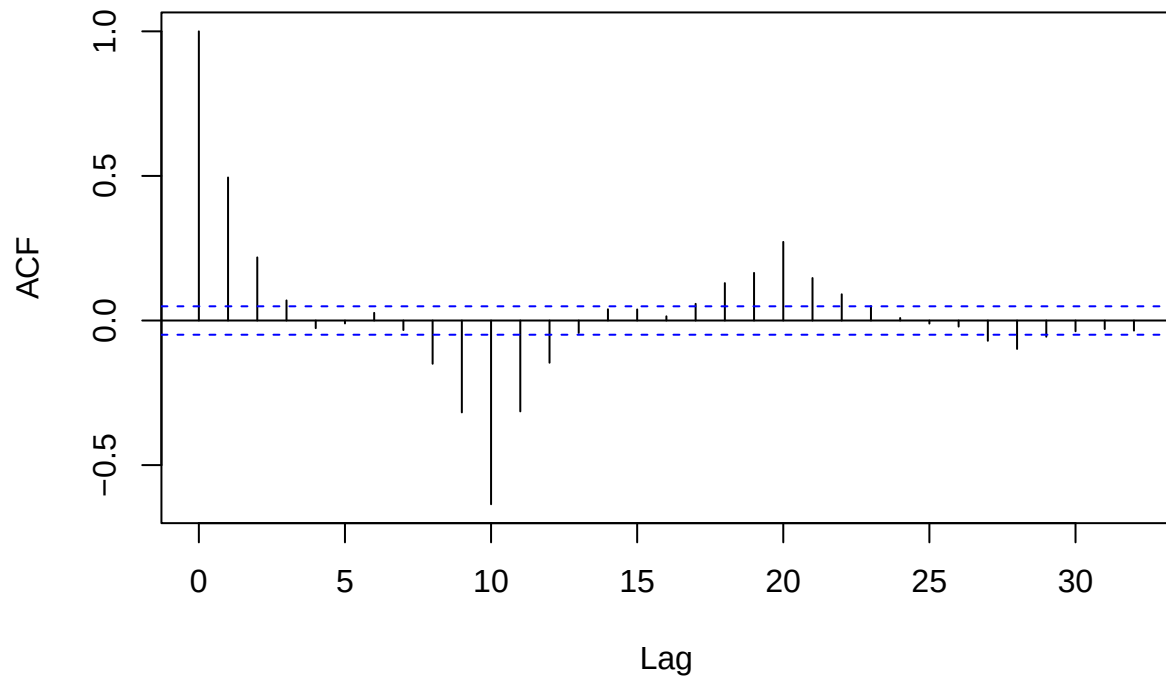


ACF and PACF of seasonally differenced series:

Seasonally Differenced Series ($s = 10$)

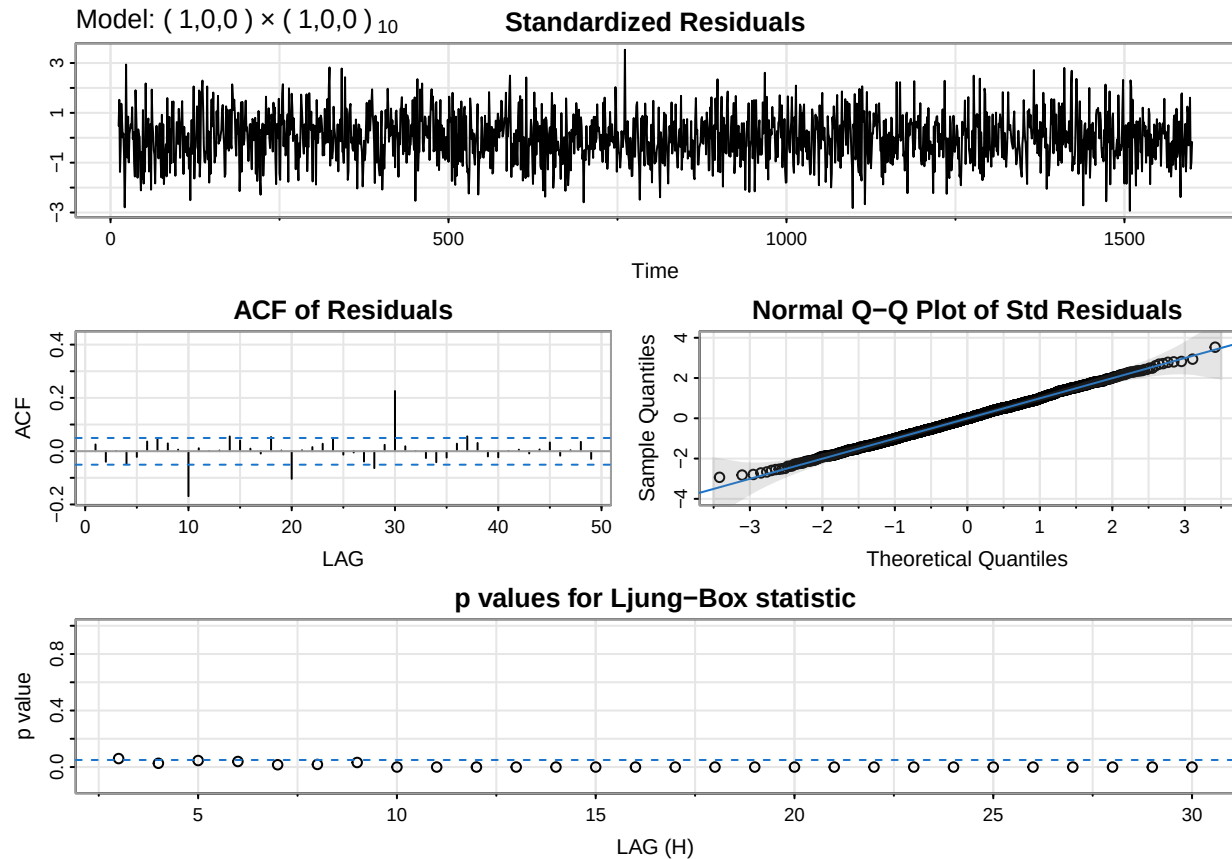


ACF of Seasonally Differenced Series ($s = 10$)



[illegible]

```
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4933 0.0218 22.6222 0.0000
## sar1     -0.6419 0.0193 -33.1945 0.0000
## xmean     0.0220 0.0319  0.6897 0.4905
##
## sigma^2 estimated as 1.112437 on 1586 degrees of freedom
##
## AIC = 2.952982  AICc = 2.952992  BIC = 2.966502
##
```



Additional models fit with three more seasonal terms than the SARIMA(1,1,0)x(1,1,0)[10]:

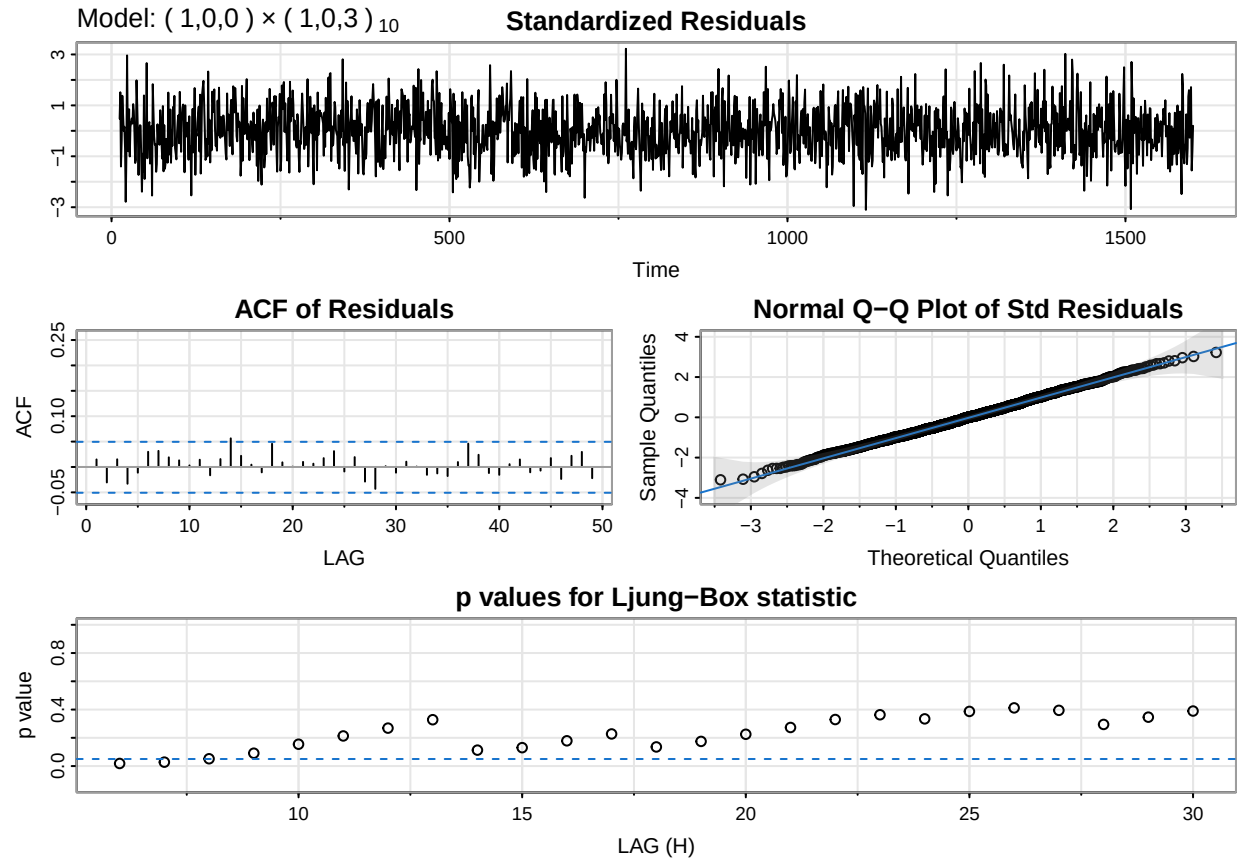
```
##
## Fitting model: SARIMA(1,0,0)(1,0,3)[10]
## initial value 0.455108
## iter 2 value 0.107986
## iter 3 value 0.038063
## iter 4 value 0.027863
## iter 5 value 0.001310
## iter 6 value 0.000687
## iter 7 value 0.000670
## iter 8 value 0.000668
## iter 9 value 0.000664
## iter 10 value 0.000652
## iter 11 value 0.000634
```

```
## iter 12 value 0.000611
## iter 13 value 0.000596
## iter 14 value 0.000588
## iter 15 value 0.000585
## iter 16 value 0.000573
## iter 17 value 0.000573
## iter 18 value 0.000573
## iter 19 value 0.000573
## iter 20 value 0.000573
## iter 21 value 0.000572
## iter 22 value 0.000571
## iter 23 value 0.000571
## iter 24 value 0.000571
## iter 25 value 0.000571
## iter 26 value 0.000571
## iter 26 value 0.000571
## iter 26 value 0.000571
## final value 0.000571
## converged
## initial value 0.004618
## iter 2 value 0.004617
## iter 3 value 0.004617
## iter 4 value 0.004617
## iter 5 value 0.004616
## iter 6 value 0.004616
## iter 7 value 0.004616
## iter 8 value 0.004616
## iter 9 value 0.004616
## iter 10 value 0.004616
## iter 11 value 0.004616
## iter 12 value 0.004616
## iter 13 value 0.004615
## iter 14 value 0.004612
## iter 15 value 0.004611
## iter 16 value 0.004610
## iter 17 value 0.004610
## iter 18 value 0.004609
## iter 19 value 0.004609
## iter 20 value 0.004609
## iter 21 value 0.004609
## iter 22 value 0.004609
## iter 23 value 0.004609
## iter 24 value 0.004609
## iter 25 value 0.004609
## iter 26 value 0.004609
## iter 27 value 0.004609
## iter 28 value 0.004609
## iter 29 value 0.004609
## iter 30 value 0.004609
## iter 31 value 0.004609
## iter 32 value 0.004609
## iter 33 value 0.004609
## iter 34 value 0.004609
## iter 35 value 0.004609
```

```

## iter 36 value 0.004609
## iter 37 value 0.004609
## iter 38 value 0.004609
## iter 39 value 0.004609
## iter 40 value 0.004609
## iter 41 value 0.004609
## iter 42 value 0.004609
## iter 43 value 0.004609
## iter 44 value 0.004609
## iter 45 value 0.004609
## iter 46 value 0.004609
## iter 47 value 0.004609
## iter 47 value 0.004609
## iter 47 value 0.004609
## final value 0.004609
## converged
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5100 0.0216 23.6020 0.0000
## sar1     -0.3176 0.8889 -0.3573 0.7209
## sma1     -0.5064 0.8904 -0.5687 0.5696
## sma2      0.1917 0.7302  0.2625 0.7930
## sma3      0.1286 0.3983  0.3228 0.7469
## xmean     0.0207 0.0316  0.6534 0.5136
##
## sigma^2 estimated as 1.003849 on 1583 degrees of freedom
##
## AIC = 2.855906  AICc = 2.855939  BIC = 2.879566
##

```

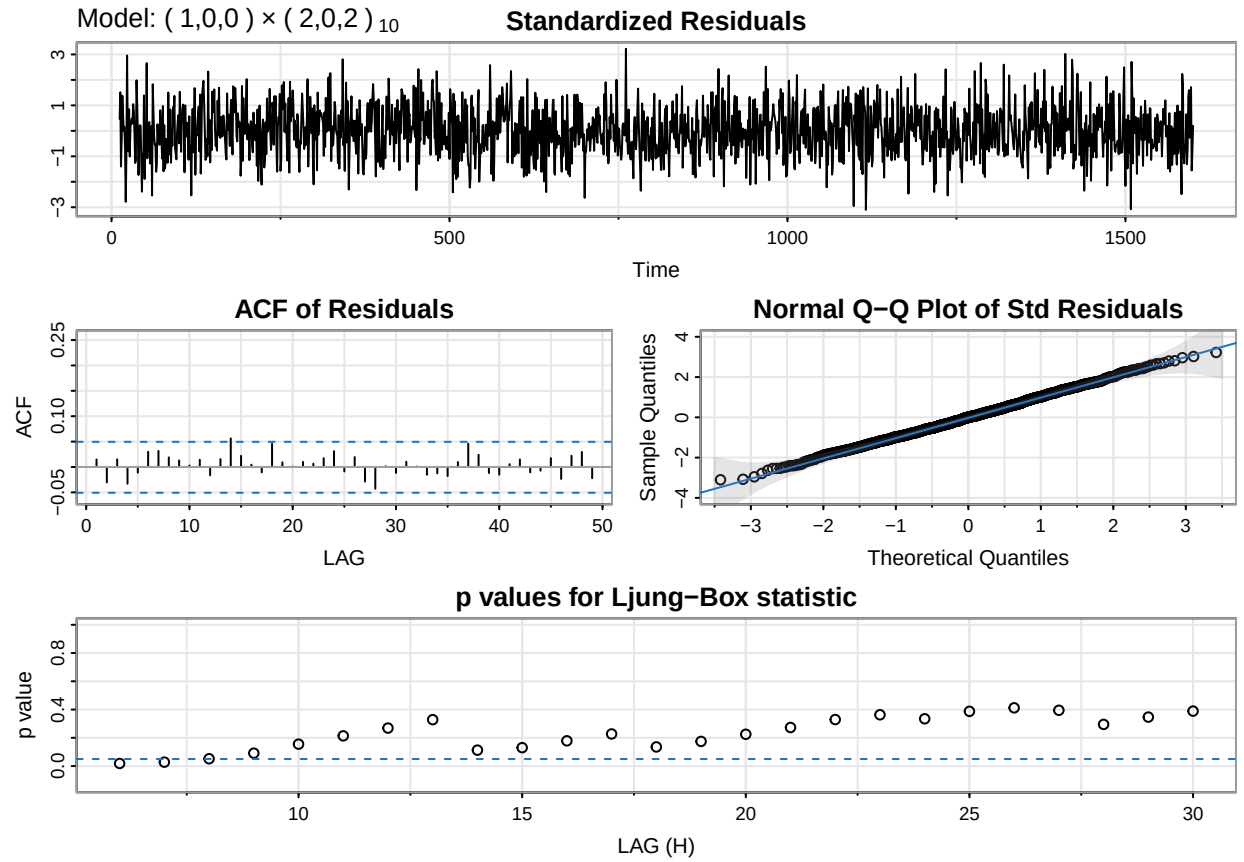


```
## AIC for SARIMA(1,0,0)(1,0,3)[10] : 4536.035
##
## Fitting model: SARIMA(1,0,0)(2,0,2)[10]
## initial value 0.454276
## iter 2 value 0.146535
## iter 3 value 0.029744
## iter 4 value 0.012560
## iter 5 value 0.007120
## iter 6 value 0.004572
## iter 7 value 0.003894
## iter 8 value 0.003828
## iter 9 value 0.003734
## iter 10 value 0.003564
## iter 11 value 0.003281
## iter 12 value 0.003094
## iter 13 value 0.002992
## iter 14 value 0.002983
## iter 15 value 0.002982
## iter 16 value 0.002979
## iter 17 value 0.002978
## iter 18 value 0.002978
## iter 19 value 0.002978
## iter 20 value 0.002978
## iter 21 value 0.002978
## iter 22 value 0.002978
## iter 23 value 0.002978
```

```

## iter 24 value 0.002978
## iter 25 value 0.002978
## iter 26 value 0.002978
## iter 26 value 0.002978
## iter 26 value 0.002978
## final value 0.002978
## converged
## initial value 0.004938
## iter 2 value 0.004892
## iter 3 value 0.004831
## iter 4 value 0.004780
## iter 5 value 0.004745
## iter 6 value 0.004682
## iter 7 value 0.004645
## iter 8 value 0.004621
## iter 9 value 0.004617
## iter 10 value 0.004617
## iter 11 value 0.004616
## iter 12 value 0.004616
## iter 13 value 0.004616
## iter 14 value 0.004616
## iter 15 value 0.004616
## iter 15 value 0.004616
## iter 15 value 0.004616
## final value 0.004616
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5100 0.0216 23.5827 0.0000
## sar1     -0.0176 0.0756 -0.2325 0.8162
## sar2      0.0081 0.0507  0.1597 0.8732
## sma1     -0.8062 0.0715 -11.2829 0.0000
## sma2      0.4305 0.0391 11.0085 0.0000
## xmean     0.0206 0.0317  0.6512 0.5150
##
## sigma^2 estimated as 1.003863 on 1583 degrees of freedom
##
## AIC = 2.855919  AICc = 2.855952  BIC = 2.879579
##

```



```
## AIC for SARIMA(1,0,0)(2,0,2)[10] : 4536.055
```

```
##
```

```
## Fitting model: SARIMA(1,0,0)(3,0,1)[10]
```

```
## initial value 0.453589
```

```
## iter 2 value 0.187751
```

```
## iter 3 value 0.036061
```

```
## iter 4 value 0.021792
```

```
## iter 5 value 0.008137
```

```
## iter 6 value 0.007646
```

```
## iter 7 value 0.007576
```

```
## iter 8 value 0.007539
```

```
## iter 9 value 0.007296
```

```
## iter 10 value 0.007008
```

```
## iter 11 value 0.006830
```

```
## iter 12 value 0.006728
```

```
## iter 13 value 0.006715
```

```
## iter 14 value 0.006714
```

```
## iter 15 value 0.006714
```

```
## iter 16 value 0.006713
```

```
## iter 17 value 0.006713
```

```
## iter 18 value 0.006713
```

```
## iter 19 value 0.006713
```

```
## iter 20 value 0.006713
```

```
## iter 21 value 0.006713
```

```
## iter 22 value 0.006713
```

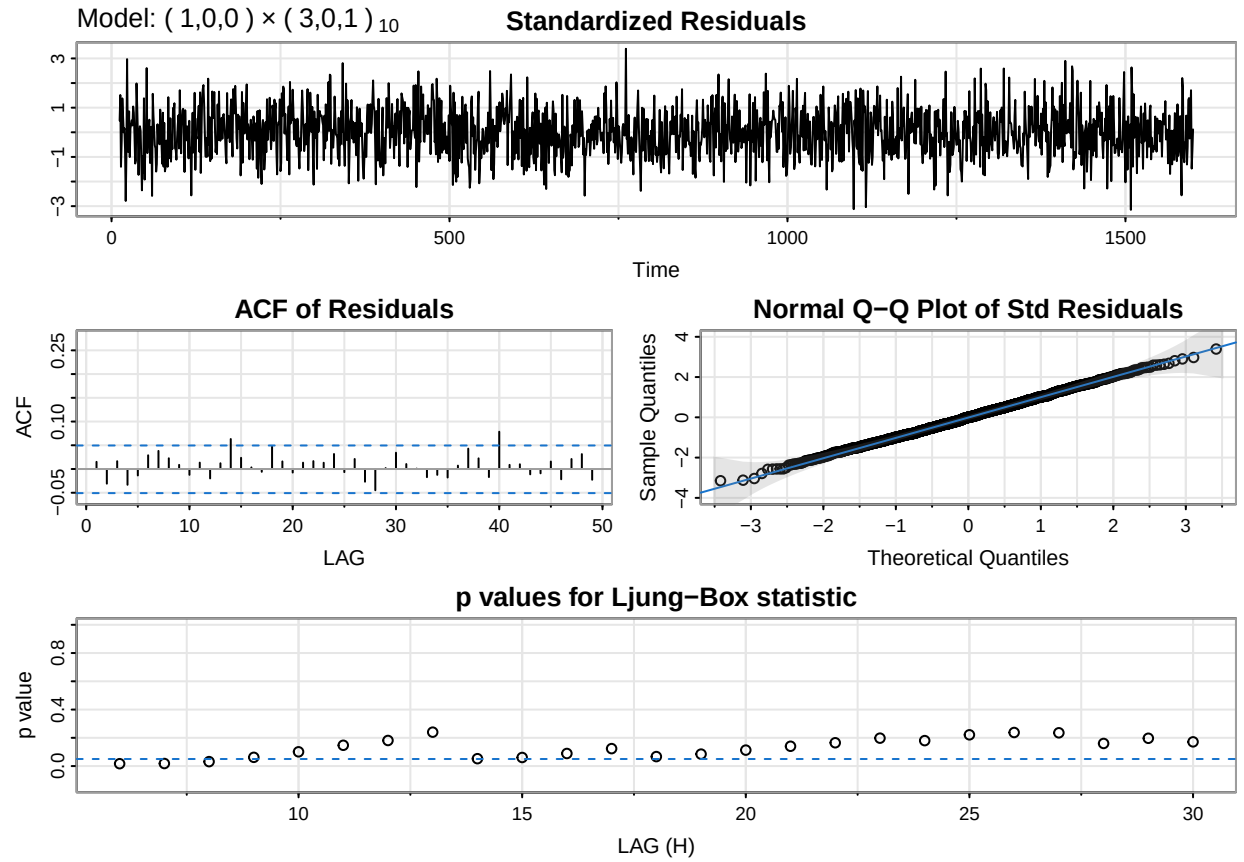
```
## iter 23 value 0.006713
```



```

## iter 24 value 0.006713
## iter 24 value 0.006713
## iter 24 value 0.006713
## final value 0.006713
## converged
## initial value 0.012273
## iter 2 value 0.012247
## iter 3 value 0.012227
## iter 4 value 0.012217
## iter 5 value 0.012173
## iter 6 value 0.012126
## iter 7 value 0.012076
## iter 8 value 0.012041
## iter 9 value 0.012026
## iter 10 value 0.012009
## iter 11 value 0.011992
## iter 12 value 0.011985
## iter 13 value 0.011983
## iter 14 value 0.011982
## iter 15 value 0.011982
## iter 16 value 0.011982
## iter 17 value 0.011982
## iter 18 value 0.011982
## iter 19 value 0.011982
## iter 20 value 0.011982
## iter 21 value 0.011982
## iter 22 value 0.011982
## iter 23 value 0.011982
## iter 24 value 0.011982
## iter 25 value 0.011982
## iter 26 value 0.011982
## iter 26 value 0.011982
## iter 26 value 0.011982
## final value 0.011982
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.5096 0.0216 23.5534 0.0000
## sar1     -0.1398 0.0992 -1.4091 0.1590
## sar2      0.3256 0.0769  4.2360 0.0000
## sar3      0.2580 0.0306  8.4411 0.0000
## sma1     -0.6618 0.0997 -6.6381 0.0000
## xmean     0.0205 0.0313  0.6539 0.5133
##
## sigma^2 estimated as 1.019393 on 1583 degrees of freedom
##
## AIC = 2.870652  AICc = 2.870685  BIC = 2.894312
##

```

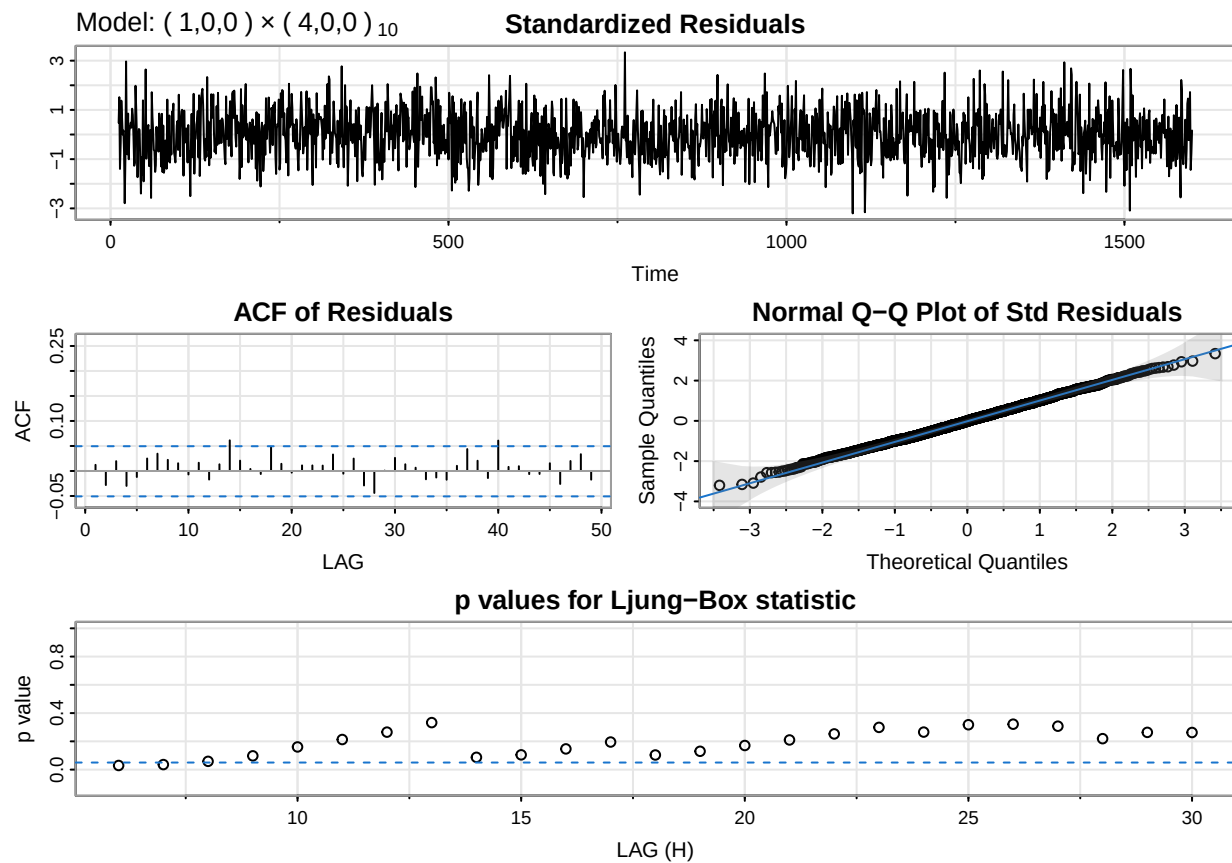


```
## AIC for SARIMA(1,0,0)(3,0,1)[10] : 4559.466
##
## Fitting model: SARIMA(1,0,0)(4,0,0)[10]
## initial value 0.451441
## iter 2 value 0.185269
## iter 3 value 0.081523
## iter 4 value 0.017359
## iter 5 value 0.005656
## iter 6 value 0.001621
## iter 7 value 0.001062
## iter 8 value 0.000755
## iter 9 value 0.000749
## iter 10 value 0.000749
## iter 11 value 0.000749
## iter 12 value 0.000749
## iter 13 value 0.000749
## iter 14 value 0.000749
## iter 15 value 0.000749
## iter 15 value 0.000749
## iter 15 value 0.000749
## final value 0.000749
## converged
## initial value 0.010900
## iter 2 value 0.010882
## iter 3 value 0.010879
## iter 4 value 0.010878
```

```

## iter    5 value 0.010877
## iter    6 value 0.010877
## iter    7 value 0.010877
## iter    8 value 0.010877
## iter    8 value 0.010877
## iter    8 value 0.010877
## final   value 0.010877
## converged
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.5077 0.0217  23.4173 0.0000
## sar1     -0.8089 0.0251 -32.1973 0.0000
## sar2     -0.2135 0.0325  -6.5775 0.0000
## sar3      0.1270 0.0326   3.8933 0.0001
## sar4      0.1087 0.0255   4.2573 0.0000
## xmean     0.0214 0.0288   0.7442 0.4568
##
## sigma^2 estimated as 1.017086 on 1583 degrees of freedom
##
## AIC = 2.868441  AICc = 2.868475  BIC = 2.892101
##

```



```

## AIC for SARIMA(1,0,0)(4,0,0)[10] : 4555.953
##

```

```
## AIC values for all models:
## $`SARIMA(1,0,0)(1,0,3)[10]`
## [1] 4536.035
##
## $`SARIMA(1,0,0)(2,0,2)[10]`
## [1] 4536.055
##
## $`SARIMA(1,0,0)(3,0,1)[10]`
## [1] 4559.466
##
## $`SARIMA(1,0,0)(4,0,0)[10]`
## [1] 4555.953
```